

Lab3 : Docker

Problem 1

Create a reactjs simple app

```
npx create-react-app myapp  
cd myapp
```

Create a dockerfile to containerize the reactapp

```
# Stage 1: Build React app  
FROM node:18 AS build  
WORKDIR /app  
COPY package*.json ./  
RUN npm install  
COPY . .  
RUN npm run build  
  
# Stage 2: Serve with nginx  
FROM nginx:alpine  
COPY --from=build /app/build /usr/share/nginx/html  
EXPOSE 80  
CMD ["nginx", "-g", "daemon off;"]
```

Build the image and test it

```
docker build -t myapp .  
docker run -d -p 3000:80 --name react-app myapp  
curl localhost:3000
```

```

nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$ vim Dockerfile
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$ docker build -t myapp .
[+] Building 43.8s (16/16) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 309B                                0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine    1.2s
=> [internal] load metadata for docker.io/library/node:18         1.2s
=> [auth] library/node:pull token for registry-1.docker.io        0.0s
=> [auth] library/nginx:pull token for registry-1.docker.io      0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [build 1/6] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d 0.0s
=> CACHED [stage-1 1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4 0.0s
=> [internal] load build context                                  2.7s
=> => transferring context: 236.85MB                                2.7s
=> CACHED [build 2/6] WORKDIR /app                                0.0s
=> [build 3/6] COPY package*.json ./                              0.5s
=> [build 4/6] RUN npm install                                    24.7s
=> [build 5/6] COPY . .                                           8.0s
=> [build 6/6] RUN npm run build                                  6.1s
=> [stage-1 2/2] COPY --from=build /app/build /usr/share/nginx/html 0.0s
=> exporting to image                                             0.0s
=> => exporting layers                                             0.0s
=> => writing image sha256:207522497f4b74aee388dcd0fc988e68ec95c9d8749b3233886ece0bbf2c4fd8 0.0s
=> => naming to docker.io/library/myapp                           0.0s
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$ docker run -d -p 3000:80 --name react-app myapp
9630c62895354e87ea27015d36b7b2404589ca7aa374db98cba1073347435d47
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED      STATUS      PORTS
9630c6289535  myapp    "/docker-entrypoint...." 38 seconds ago Up 38 seconds 0.0.0.0:3000->80/tcp, [::]:3000->80/tc
p react-app
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$ localhost:3000
localhost:3000: command not found
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3/myapp$ curl localhost:3000
<!doctype html><html lang="en"><head><meta charset="utf-8"/><link rel="icon" href="/favicon.ico"/><meta name="viewport" c
ontent="width=device-width,initial-scale=1"/><meta name="theme-color" content="#000000"/><meta name="description" content
="Web site created using create-react-app"/><link rel="apple-touch-icon" href="/logo192.png"/><link rel="manifest" href=
"/manifest.json"/><title>React App</title><script defer="defer" src="/static/js/main.a8e766ca.js"></script><link href="/st
atic/css/main.f855e6bc.css" rel="stylesheet"></head><body><noscript>You need to enable JavaScript to run this app.</noscr

```

Problem 2

Create a dockerfile for ubuntu image which sleeps by default for 5 sec or sleeps according to the given number in the docker command

```

FROM ubuntu:latest
CMD ["sh", "-c", "sleep ${SLEEP_TIME:-5}"]

```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker login
Authenticating with existing credentials... [Username: mnagy156]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker tag myapp mnagy156/myapp:latest
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker tag sleep-ubuntu mnagy156/sleep-ubuntu:latest
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker push mnagy156/myapp:latest
The push refers to repository [docker.io/mnagy156/myapp]
4896866dde21: Pushed
da3ac26fdf0f: Mounted from library/nginx
7b2903554e63: Mounted from library/nginx
6c0e59fd138a: Mounted from library/nginx
aa0fc249df10: Mounted from library/nginx
f35bcec50d8c: Mounted from library/nginx
660f9a93104f: Mounted from library/nginx
53998a5033c3: Mounted from library/nginx
989e799e6349: Mounted from library/nginx
latest: digest: sha256:28a4748ca67b6a0ef1b3f6d788e6f4bd04ba2defac6f4080912731cdd66b2411 size: 2199
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker push mnagy156/sleep-ubuntu:latest
The push refers to repository [docker.io/mnagy156/sleep-ubuntu]
efafae78d70c: Mounted from library/ubuntu
latest: digest: sha256:9c8977d3b8df2610add23e198e4d8b34d332e49c693be7b88a51443f3e65129b size: 529
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
```

Problem 3

Push the images created in Problem 1&2 into your docker hub repo

```
docker login
docker tag myapp mnagy156/myapp:latest
docker tag sleep-ubuntu mnagy156/sleep-ubuntu:latest
docker push mnagy156/myapp:latest
docker push mnagy156/sleep-ubuntu:latest
```

```

nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker login
Authenticating with existing credentials... [Username: mnagy156]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker tag myapp mnagy156/myapp:latest
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker tag sleep-ubuntu mnagy156/sleep-ubuntu:latest
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker push mnagy156/myapp:latest
The push refers to repository [docker.io/mnagy156/myapp]
4896866dde21: Pushed
da3ac26fdf0f: Mounted from library/nginx
7b2903554e63: Mounted from library/nginx
6c0e59fd138a: Mounted from library/nginx
aa0fc249df10: Mounted from library/nginx
f35bcec50d8c: Mounted from library/nginx
660f9a93104f: Mounted from library/nginx
53998a5033c3: Mounted from library/nginx
989e799e6349: Mounted from library/nginx
latest: digest: sha256:28a4748ca67b6a0ef1b3f6d788e6f4bd04ba2defac6f4080912731cdd66b2411 size: 2199
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker push mnagy156/sleep-ubuntu:latest
The push refers to repository [docker.io/mnagy156/sleep-ubuntu]
efafae78d70c: Mounted from library/ubuntu
latest: digest: sha256:9c8977d3b8df2610add23e198e4d8b34d332e49c693be7b88a51443f3e65129b size: 529
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$

```

Problem 4

Create 2 nginx containers with network type bridge, enter to one of them and use curl command to view the content of the other container.

```

docker network create my-net
docker run -d --name nginx1 --network my-net nginx
docker run -d --name nginx2 --network my-net nginx
docker network ls | grep my-net
docker exec -it nginx1 sh
curl http://nginx2

```

```

nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker network create my-net
48a0f23ecd522398141c19d975de34dele18f6dc6885467c10bfd4b35eb95a35
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker run -d --name nginx1 --network my-net nginx
8da7c7b3b50d619ddd7e2b366f7ea16a59706766e0c03effadc592dac6c98713
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker run -d --name nginx2 --network my-net nginx
0c0abeaa30b651eb7e73b1607a0860a62b52f8a719cce9481b196832bbf26f13
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker network ls | grep my-net
48a0f23ecd52 my-net bridge local
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$

```

```
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$
nagy@ZenBook-UX425EA:~/Mine/ITI/016_Docker/Solutions/Lab3$ docker exec -it nginx1 sh
#
# curl http://nginx2
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
#
# █
```